

LUCIDMSK CME PROGRAM

MODULE 04 — POST-TEST

Rheumatologic Arthropathies: Pattern-Based Radiographic Diagnosis

General MSK · 10 Questions · 1.0 AMA PRA Cat 1 Credit(s)[™]

Read each question carefully and select the single best answer. A score of $\geq$ **70%** (7/10 correct) is required to claim CME credit. The complete answer key with detailed explanations begins after Question 10.

POST-TEST QUESTIONS

1 Symmetric MCP erosions, periarticular osteopenia, pancarpal loss, DIP sparing. Diagnosis:

- A Psoriatic arthritis
- B Erosive OA
- C Rheumatoid arthritis
- D Gout

2 Overhanging edge erosion with sclerotic margin and PRESERVED joint space is pathognomonic for:

- A CPPD
- B RA
- C Gout
- D PsA

3 Asymmetric sacroiliitis + dactylitis ("sausage digit"). Most likely diagnosis:

- A Ankylosing spondylitis
- B Psoriatic arthritis
- C RA
- D CPPD

4 CPPD is most sensitively detected on wrist radiograph by calcification of which structure?

- A Scapholunate ligament
- B Triangular fibrocartilage complex (TFC)
- C Flexor carpi radialis tendon
- D ECU tendon

5 ADI measures 5 mm in an adult RA patient. Clinical significance:

- A Normal finding
- B Surgical emergency
- C Atlantoaxial subluxation — neurosurgical referral required
- D No follow-up needed

6 Delicate marginal vertical syndesmophytes + BILATERAL SYMMETRIC sacroiliitis. Diagnosis:

- A PsA
- B Ankylosing spondylitis
- C DISH
- D Reactive arthropathy

7 DIP erosions; pencil-in-cup deformity; preserved bone density; asymmetric. Diagnosis:

- A RA
- B Erosive OA
- C Psoriatic arthritis
- D Gout

8 CPPD knee preferentially narrows which compartment compared to OA?

- A Medial tibiofemoral
- B Lateral tibiofemoral
- C Patellofemoral
- D Posterior compartment

9 Bilateral sclerosis and erosions with PARTIAL ankylosis but NOT complete fusion. Modified New York sacroiliitis grade:

A	Grade 1
B	Grade 2
C	Grade 3
D	Grade 4

10

Triangular iliac-sided sclerosis at the SI joint bilaterally; NO erosions; 28-year-old postpartum woman:

A	Ankylosing spondylitis
B	Osteitis condensans ilii (OCI)
C	Septic sacroiliitis
D	PsA

ANSWER KEY & EXPLANATIONS

Q	1	2	3	4	5	6	7	8	9	10
ANS	C	C	B	B	C	B	C	C	C	B

Q1 Symmetric MCP erosions, periarticular osteopenia, pancarpal loss, DIP sparing. Diagnosis: **Answer: C**

A Psoriatic arthritis

B Erosive OA

✓ **Rheumatoid arthritis**

D Gout

EXPLANATION RA: symmetric + periarticular osteopenia + marginal MCP erosions + pancarpal + DIP sparing. PsA has DIP involvement. Gout has overhanging edge erosions. EOA affects DIPs centrally.

Q2 Overhanging edge erosion with sclerotic margin and **PRESERVED** joint space is pathognomonic for: **Answer: C**

A CPPD

B RA

✓ **Gout**

D PsA

EXPLANATION Overhanging (rat-bite) erosions with sclerotic borders at a distance from the joint margin, with preserved joint space = pathognomonic gout. All other arthropathies narrow the joint space before extensive erosion.

Q3 Asymmetric sacroiliitis + dactylitis ("sausage digit"). Most likely diagnosis: **Answer: B**

A Ankylosing spondylitis

✓ **Psoriatic arthritis**

C RA

D CPPD

EXPLANATION PsA = asymmetric sacroiliitis + dactylitis + DIP involvement + ray distribution. AS causes symmetric bilateral sacroiliitis. RA does not cause sacroiliitis or dactylitis.

Q4 CPPD is most sensitively detected on wrist radiograph by calcification of which structure? **Answer: B**

A Scapholunate ligament

✓ **Triangular fibrocartilage complex (TFC)**

C Flexor carpi radialis tendon

D ECU tendon

EXPLANATION CPPD most sensitively involves the TFC on PA wrist radiograph — chondrocalcinosis here is the most sensitive single radiographic sign of CPPD.

Q5 ADI measures 5 mm in an adult RA patient. Clinical significance:

Answer: C

- A Normal finding
- B Surgical emergency

✓ **Atlantoaxial subluxation — neurosurgical referral required**

- D No follow-up needed

EXPLANATION ADI >3 mm in adults = atlantoaxial subluxation (RA pannus erodes transverse ligament). ADI >9 mm = spinal cord injury risk. Surgical referral is required but not automatically emergent without neurological signs.

Q6 Delicate marginal vertical syndesmophytes + BILATERAL SYMMETRIC sacroiliitis. Diagnosis:

Answer: B

- A PsA

✓ **Ankylosing spondylitis**

- C DISH
- D Reactive arthropathy

EXPLANATION AS = delicate marginal vertical syndesmophytes (bamboo spine) + symmetric bilateral sacroiliitis. PsA/ReA = asymmetric SI + bulky non-marginal ossifications. DISH = flowing anterior osteophytes; no sacroiliitis.

Q7 DIP erosions; pencil-in-cup deformity; preserved bone density; asymmetric. Diagnosis:

Answer: C

- A RA
- B Erosive OA

✓ **Psoriatic arthritis**

- D Gout

EXPLANATION PsA = DIP erosions + pencil-in-cup + preserved density + asymmetric + ray distribution. EOA also affects DIPs but shows central "seagull" erosion + osteophytes. RA spares DIPs.

Q8 CPPD knee preferentially narrows which compartment compared to OA?

Answer: C

- A Medial tibiofemoral
- B Lateral tibiofemoral

✓ **Patellofemoral**

- D Posterior compartment

EXPLANATION CPPD preferentially narrows the PATELLOFEMORAL compartment — unusual distribution that flags the diagnosis. OA preferentially narrows the medial tibiofemoral compartment.

Q9	Bilateral sclerosis and erosions with PARTIAL ankylosis but NOT complete fusion. Modified New York sacroiliitis grade:	Answer: C
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A Grade 1

B Grade 2

✓ **Grade 3**

D Grade 4

EXPLANATION Grade 3 = erosion + sclerosis + widening/narrowing + partial ankylosis. Grade 2 = small erosions/sclerosis without joint space change. Grade 4 = complete ankylosis.

Q10	Triangular iliac-sided sclerosis at the SI joint bilaterally; NO erosions; 28-year-old postpartum woman:	Answer: B
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A Ankylosing spondylitis

✓ **Osteitis condensans ilii (OCI)**

C Septic sacroiliitis

D PsA

EXPLANATION OCI = triangular iliac-sided sclerosis WITHOUT erosions, in childbearing-age women — benign and self-limiting. AS shows erosions and progresses to fusion. Septic = unilateral + acute symptoms.